

## Predicting Sales for Different Stores

The goal of this project is to build a machine learning model that predicts the total sales for different stores based on various factors, such as store status, promotions, holidays, and customer visits. The model aims to help businesses optimize operations by forecasting sales for future days.

There are 1 million records in this dataset.

You can use the first 700k records for building the model, 200k records for evaluating the model, and the rest of the records as live records on which the model would be evaluated in production.

Please create separate .py files for model training and prediction. Make sure that your dataset during the model building and prediction phases goes through the same data processing and feature creation steps.

Your final deliverable can either be a Python script, which can be scheduled as a batch script to process sales predictions in batch, or a Flask API endpoint, which can be used to predict sales in real time.

Please submit the final report, which should give a clear idea on the following points:

- ML model selected.
- Features selected.
- Feature importance.
- Evaluation of the model.
- Original code.

**Use the "predicting\_sales" table from the Regression.db file for the project.**

Please feel free to reach out to your mentor in case of any doubts.

### Data Dictionary -

Store - The unique identifier for each store.

DayOfWeek - Represents the day of the week as a numerical value.  
(0 = Monday, 6 = Sunday).

Date - The calendar date for the sales record in the format YYYY-MM-DD.

Sales - The total revenue (in monetary units) generated by the store on a specific day.

Customers - The total number of customers who visited the store on a specific day.

Open - A binary indicator showing whether the store was open (1) or closed (0) on a specific day.

Promo - A binary indicator showing whether the store was running a promotion (1) or not (0) on a specific day.

StateHoliday - Denotes whether the day was a public holiday (e.g., 'a', 'b', 'c', or '0' for no holiday).

SchoolHoliday - A binary indicator showing whether the day was a school holiday (1) or not (0).